

INDEPENDENCE, IA, USA

WMO: 720293

Lat: 42.453N

Lon: 91.948W

Elev: 978

StdP: 14.18

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04989

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.7	-4.1	-15.0	2.5	-8.8	-9.3	3.5	-3.3	32.1	24.7	28.4	21.1	7.4	310	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	90.1	74.9	87.6	73.6	85.5	71.6	77.4	86.4	75.4	84.2	73.5	81.5	10.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.6	134.4	83.2	72.6	125.1	80.4	71.6	120.9	79.1	41.7	85.7	39.4	84.3	37.8	81.3	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.9	24.1	20.6	-18.8	96.1	5.8	7.0	-23.0	101.1	-26.4	105.2	-29.7	109.2	-34.0	114.3		
(5)			-19.1	81.2	5.8	1.8	-23.3	82.5	-26.7	83.6	-29.9	84.6	-34.1	85.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.7	19.3	22.1	35.9	47.8	60.0	69.4	72.0	69.2	63.4	49.8	36.8	25.3
	DBStd	20.85	12.51	12.07	11.45	9.69	8.86	5.78	5.57	5.44	7.96	9.28	10.17	11.51
	HDD50	3659	952	782	457	153	18	0	0	0	5	118	408	766
	HDD65	6974	1417	1202	902	520	204	22	8	20	124	478	846	1231
	CDD50	2827	0	0	20	86	330	581	682	597	406	113	12	0
	CDD65	666	0	0	1	2	50	153	225	152	75	8	0	0
	CDH74	6102	0	0	14	75	546	1368	2038	1271	701	89	0	0
	CDH80	1854	0	0	0	16	162	427	688	343	202	16	0	0
Wind	WSAvg	9.4	10.7	10.8	11.3	12.3	11.0	8.7	6.3	5.6	7.1	8.7	10.5	10.3
Precipitation	PrecAvg	38.3	1.1	1.5	2.1	3.7	5.1	5.6	4.8	4.4	2.9	2.6	2.3	1.5
	PrecMax	51.0	2.4	4.5	5.3	8.4	10.7	9.1	12.8	10.1	5.0	6.4	5.8	3.0
	PrecMin	29.3	0.4	0.2	0.2	1.1	0.9	0.9	0.8	0.8	0.3	0.6	0.4	0.3
	PrecStd	6.1	0.5	1.0	1.4	1.9	2.1	2.1	2.8	2.6	1.2	1.7	1.6	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.1	58.2	76.7	81.7	90.7	92.0	93.3	90.3	90.4	81.6	69.5	54.7
		MCWB	41.9	49.9	61.0	61.8	70.8	74.0	77.7	74.4	72.8	63.5	56.5	49.3
	2%	DB	41.2	45.9	66.2	74.7	84.8	88.3	90.1	87.6	86.1	76.6	61.2	47.7
		MCWB	37.7	42.3	55.8	59.1	67.1	72.5	77.4	74.3	70.6	61.7	51.3	43.3
	5%	DB	36.8	41.0	59.2	70.4	79.5	85.9	86.7	84.0	81.6	71.6	55.2	43.5
		MCWB	34.8	37.4	51.7	56.0	63.4	69.9	74.9	72.6	67.7	60.6	48.3	39.9
	10%	DB	34.2	36.2	52.2	64.4	76.6	81.4	83.9	81.1	77.8	66.1	52.0	38.4
		MCWB	33.0	34.1	46.5	53.6	62.9	67.9	72.8	70.1	65.0	57.9	46.6	36.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.4	51.4	62.2	64.0	72.3	78.2	80.8	78.5	76.2	69.8	58.8	50.4
		MCDB	47.2	58.2	73.8	76.9	85.5	87.6	89.2	86.7	85.5	74.7	64.5	52.4
	2%	WB	38.0	42.4	57.4	60.9	69.2	75.2	78.4	76.2	73.2	66.1	54.0	45.5
		MCDB	41.0	46.0	64.0	72.0	80.5	84.4	86.8	84.4	81.7	71.7	59.8	46.7
	5%	WB	35.0	37.6	51.8	58.2	67.0	72.7	76.2	74.0	70.7	61.8	50.2	40.7
		MCDB	36.6	40.8	58.1	67.8	76.8	81.1	85.1	81.4	77.6	67.5	55.3	43.9
	10%	WB	32.9	34.1	46.8	54.8	64.4	70.7	74.2	72.2	68.5	57.9	46.3	36.6
		MCDB	33.9	35.6	52.2	63.0	72.0	78.0	82.4	78.8	75.0	65.4	50.4	38.4
Mean Daily Temperature Range	5% DB	MDBR	15.8	16.8	17.8	21.3	21.7	20.5	20.8	21.3	23.6	21.3	18.6	14.6
		MCDBR	17.2	21.1	27.9	30.8	26.3	24.6	22.9	23.1	26.2	28.6	25.9	18.6
		MCWBR	14.0	16.2	17.3	17.2	12.0	11.5	11.8	12.2	12.4	15.0	17.1	14.7
	5% WB	MCDBR	16.2	20.9	25.7	25.8	23.0	20.8	21.3	20.0	21.0	21.8	24.5	17.5
		MCWBR	13.4	16.3	16.7	16.0	12.1	11.9	12.1	11.6	11.2	15.0	17.4	14.5
Clear-Sky Solar Irradiance	taub	0.281	0.304	0.330	0.373	0.403	0.419	0.432	0.412	0.382	0.336	0.297	0.279	
	taud	2.385	2.326	2.379	2.288	2.268	2.269	2.217	2.281	2.331	2.456	2.489	2.421	
	Ebn at Noon	275	285	290	284	277	272	266	268	269	270	267	266	
	Edh at Noon	25	31	34	40	42	42	44	40	35	28	23	22	
All-Sky Solar Radiation	RadAvg	580	904	1212	1479	1760	1940	1963	1711	1418	924	621	459	
	RadStd	58	82	125	127	122	167	121	112	106	116	49	38	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page